

Data Center Availability

Calculation Methodology & Topology Comparison (Tier I–IV)

American Terawatt

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Agenda

Methodology

Tier I & II (baseline)

Tier III ($N+1$)

Tier IV ($2N$)

Comparison & Conclusions

Methodology

What “availability” means

Availability is the fraction of time a facility (or system) can deliver IT load without interruption, measured over a defined period (typically a rolling 12 months).

Core relationship

$$A = \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

where **MTBF** is mean time between failures and **MTTR** is mean time to repair/recover.
Reported as a percentage or in “nines.”

The language of nines

Availability	Nines	Allowable downtime / yr
99 %	two	≈ 3.65 days
99.9 %	three	≈ 8.77 hours
99.99 %	four	≈ 52.6 minutes
99.999 %	five	≈ 5.26 minutes
99.9999 %	six	≈ 31.5 seconds

Every additional nine cuts allowable downtime by a factor of ten.

Combining components

A facility is modeled as a reliability block diagram (RBD) of components in *series* and *parallel*.

Series — all required

$$A = \prod_i A_i$$

Any element fails $\Rightarrow$ outage. Always *below* the weakest link.

Parallel — redundant

$$A = 1 - \prod_i (1 - A_i)$$

Survives if ≥ 1 path works. Redundancy raises A sharply.

$N+1$ unit group (tolerates one failure)

$$A_{\text{grp}} = A^{N+1} + (N+1) A^N (1 - A)$$

Modeling assumptions

Per-unit availabilities (illustrative, single-unit

MTBF/MTTR)

Component	A_{unit}
Utility feed	0.99900
Generator	0.99500
UPS module	0.99950
MV switchgear	0.99999
LV switchgear	0.99998
PDU	0.99995
Busway	0.99999
Chiller	0.99500
Pump	0.99900
CRAH unit	0.99900

Frameworks

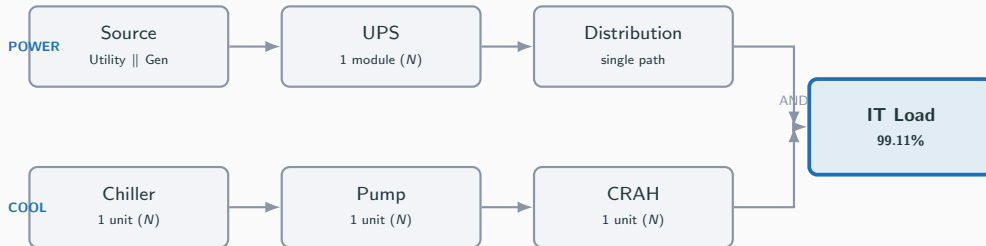
- Uptime Institute Tier standard (I–IV)
- TIA-942 infrastructure standard
- RBD, FMEA, Monte Carlo for complex topologies

Scope

- Facility *infrastructure* availability
- Planned maintenance excluded (concurrently maintainable)
- Facility = Power path $\times$ Cooling path

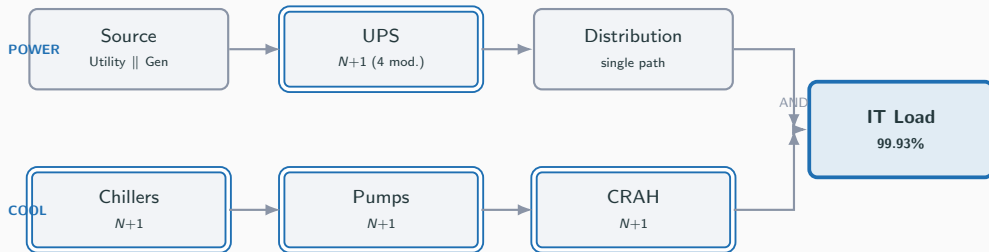
Tier I & II (baseline)

Tier I topology — N , single path (reference)



No redundancy anywhere; single distribution path. *Not* concurrently maintainable — planned maintenance requires a shutdown. Model $\approx 99.11\%$ (≈ 78 h/yr); Uptime Institute reference 99.671% (≈ 28.8 h/yr).

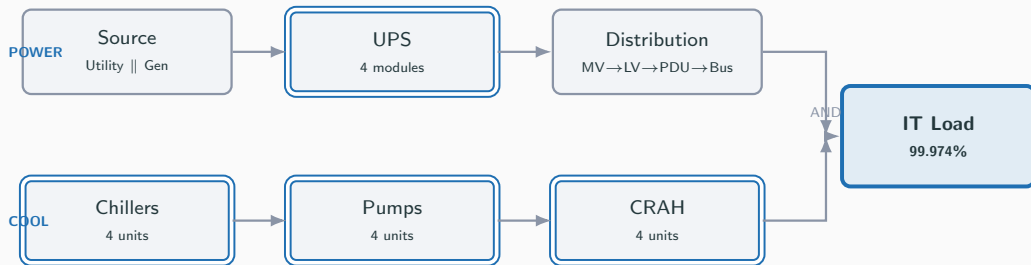
Tier II topology — $N+1$ components, single path (reference)



Redundant capacity components ($N+1$) but a single, non-redundant distribution path $\Rightarrow$ *not* concurrently maintainable. Same component RBD as Tier III; the difference is planned-maintenance downtime. Model $\approx$ 99.93 % (≈ 6.3 h/yr); Uptime Institute reference 99.741 % (≈ 22 h/yr).

Tier III (N+1)

Tier III topology — $N+1$, concurrently maintainable



- double border = $N+1$ redundant stage
- single border = single (concurrently maintainable) path.

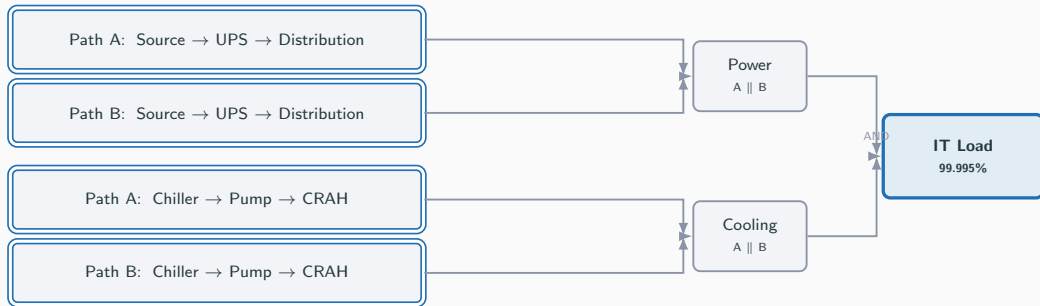
Tier III results

Stage	Availability	Downtime / yr
Source (Utility Gen)	99.9995 %	2.6 min
UPS ($N+1$)	99.9998 %	0.8 min
Distribution (series)	99.9910 %	47 min
Power path	99.9904 %	51 min
Chillers ($N+1$)	99.9851 %	78 min
Pumps ($N+1$)	99.9994 %	3 min
CRAH ($N+1$)	99.9994 %	3 min
Cooling path	99.9839 %	85 min
Facility = Power $\times$ Cooling	99.974 %	$\approx$ 135 min (2.26 h)

The single distribution path is the weakest link, capping the power path at $\approx$ four nines.

Tier IV (2N)

Tier IV topology — $2N$, fault tolerant



Every stage duplicated end-to-end ($2N$) — no single non-redundant link; unavailable only if *both* A and B fail.

Tier IV results

Subsystem	Availability	Downtime / yr
Power — per side (series string)	99.8905 %	—
Power — $2N$ ($A \parallel B$)	99.99988 %	< 1 min
Cooling — per side (series string)	99.3011 %	—
Cooling — $2N$ ($A \parallel B$)	99.99512 %	26 min
Facility = Power $\times$ Cooling	99.995 %	$\approx$ 26 min

Paralleling two full paths, $A = 1 - (1 - A_{\text{path}})^2$, *squares* each side's unavailability. Cooling is now the limiting path (single-side chillers are the weakest string).

Comparison & Conclusions

Four-tier reference

	Redundancy	Conc. maint.	Model A (downtime)	Uptime ref.
Tier I	N , single path	no	99.11 % (≈ 78 h)	99.671 % (28.8 h)
Tier II	$N+1$ comp., single path	no	99.93 % (≈ 6.3 h)	99.741 % (22 h)
Tier III	$N+1$, redundant path	yes	99.974 % (≈ 135 min)	99.982 % (1.6 h)
Tier IV	$2N$, fault tolerant	yes	99.995 % (≈ 26 min)	99.995 % (26.3 min)

Model — our per-unit component RBD. Tiers I–II add assumed planned-maintenance shutdowns (12 h / 4 h); III–IV exclude planned work.

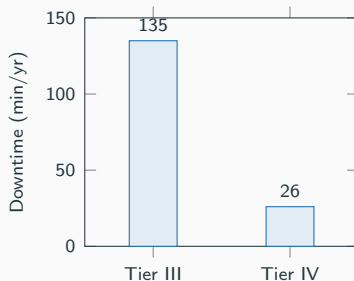
Uptime ref. — Uptime Institute's widely-cited legacy figures. They are design classifications, not measured values; the Institute has since de-emphasised specific percentages.

Tier III vs. Tier IV

	Tier III ($N+1$)	Tier IV ($2N$)
Redundancy	unit-level	full path
Distribution	single path	dual A/B
Fault tolerant	no	yes
Availability	99.974 %	99.995 %
Downtime/yr	≈ 135 min	≈ 26 min

Result

$2N$ delivers $\approx 5\times$ less downtime.



- **Illustrative values.** Per-unit figures are industry-typical, not site-measured; swap in real MTBF/MTTR before use.
- **Human error & common-cause failures dominate** real outages — calculated availability is optimistic versus observed.
- **Define “down” explicitly.** Total loss vs. partial capacity vs. SLA breach; dual-corded (A/B) IT loads complicate single-path loss.
- **Infrastructure $\neq$ IT service availability.** State which layer is measured.
- **Planned maintenance excluded** — the point of concurrent maintainability (Tier III+).

Takeaways

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}; \text{ build up from an RBD.}$$

Series multiplies (lowers); parallel/redundancy squares unavailability (raises).

Tier III $\approx 99.974\%$ $\rightarrow$ Tier IV $\approx 99.995\%$: $\approx 5\times$ less downtime.

The weakest non-redundant link sets the ceiling.